

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE															
Regular & Supplementary Winter Examination-2023															
Course: B. Tech.		Branch: Civil & Allied Engineering													
Semester: V															
Subject Code & Name: BTCVPE506A/BTCIPE506A Advanced Environmental Engineering															
Max Marks: 60		Date: 12/01/2024													
		Duration: 3 Hr.													
Instructions to the Students:															
1. All the questions are compulsory.															
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.															
3. Use of non-programmable scientific calculators is allowed.															
4. Assume suitable data wherever necessary and mention it clearly.															
		(Level/CO)	Marks												
Q. 1	Solve Any Two of the following.		12												
A)	Explain working of following with neat sketch i. Aerobic oxidation pond ii. Facultative oxidation pond	CO 1	6												
B)	Using following data Compute DO at the end of 1 day <table border="1"><tr><td></td><td>River</td><td>Wastewater</td></tr><tr><td>Flow (m³/sec)</td><td>25</td><td>3</td></tr><tr><td>DO (mg/lit)</td><td>9.1</td><td>0</td></tr><tr><td>BOD₅ (mg/lit)</td><td>2</td><td>300</td></tr></table> Take deoxygenation constant as 0.1 /day and reoxygenation constant as 0.3 /day (both are to the base 10). Take DO _{sat} = 9.10 mg/lit.		River	Wastewater	Flow (m ³ /sec)	25	3	DO (mg/lit)	9.1	0	BOD ₅ (mg/lit)	2	300	CO 1	6
	River	Wastewater													
Flow (m ³ /sec)	25	3													
DO (mg/lit)	9.1	0													
BOD ₅ (mg/lit)	2	300													
C)	Explain self-purification of stream and its process.	CO 1	6												
Q.2	Solve Any Two of the following.		12												
A)	Explain concepts Waste volume reduction with proper examples.	CO 2	6												
B)	Differentiate between Sewage and industrial wastewater.	CO 2	6												
C)	Discuss 'Control of industrial pollution'	CO 2	6												
Q. 3	Solve Any Two of the following.		12												
A)	Differentiate between Nitrification and Denitrification of wastewater.	CO 3	6												
B)	Explain working of i. SBR ii. MBBR	CO 3	6												
C)	Write suitable wastewater treatment technologies to avoid excessive growth of algae.	CO 3	6												

Q.4	Solve Any Two of the following.		12
A)	Explain chain of disease transmission.	CO 4	6
B)	Explain Life cycle of Mosquito. Also examine mosquito control w.r.t. different stages of life cycle.	CO 4	6
C)	Which are common Air borne diseases. Explain any two. Write their general symptoms.	CO 4	6
Q. 5	Solve Any Two of the following.		12
A)	Explain water quality criteria prescribed by Central Pollution Control Board (CPCB).	CO 5	6
B)	Compare Control measures for Mosquitoes and rodents in details.	CO 5	6
C)	Discuss main features and provisions under Swachh Bharat Abhiyan (Clean India Mission)	CO 5	6
	*** End ***		

The grid and the borders of the table will be hidden before final printing.